

ECON 5345 Homework 2 Report

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February 19, 2026

Question 1

a. Taking logarithm, we have

$$\log A_t = z_t + \alpha \log \Gamma_t.$$

Taking difference, we get

$$\Delta a_t = \log A_t - \log A_{t-1} = (\rho_z - 1)z_{t-1} + \alpha g_t + e_t^z.$$

Since $\{z_t\}$ and $\{g_t\}$ are stationary and $e_t^z \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_z^2)$, we have Δa_t being stationary. So $d = 1$.

Note that

$$(1 - \rho_z L)\Delta z_t = (1 - L)e_t^z, \text{ and } (1 - \rho_g L)g_t = (1 - \rho_g)\mu_g + e_t^g.$$

Multiplying $\Delta a_t = \Delta z_t + \alpha \Delta g_t$ by $(1 - \rho_z L)(1 - \rho_g L)$, we get

$$(1 - \rho_z L)(1 - \rho_g L)\Delta a_t = (1 - \rho_g L)(1 - L)e_t^z + \alpha(1 - \rho_z L)[(1 - \rho_g)\mu_g + e_t^g]. \quad (1)$$

The LHS characterizes an $AR(2)$ process and the RHS characterizes a $MA(2)$ process. So $p = 2, q = 2$.

b. Consider the stationary z part first. We have

$$z_{t+h} = \sum_{j=0}^{+\infty} \rho_z^j e_{t+h-j}^z.$$

Given $\{z_\tau\}_{\tau=-\infty}^{t-1}$, the forecast error variance is

$$FEV_h^z = \text{Var} \left[\sum_{j=0}^h \rho_z^j e_{t+h-j}^z \right] = \sigma_z^2 \frac{1 - \rho_z^{2(h+1)}}{1 - \rho_z^2}.$$

Now consider the trend component $\log \Gamma_t$. Note that

$$\log \Gamma_{t+h} = \log \Gamma_{t-1} + \sum_{j=0}^h g_{t+j}.$$

Ignoring the constant term, the g_t process can be written as

$$g_{t+h} = \sum_{j=0}^{+\infty} \rho_g^j e_{t+h-j}^g.$$

Given $\{z_\tau\}_{\tau=-\infty}^{t-1}$, the forecast error variance is

$$FEV_h^\Gamma = \text{Var} \left[\sum_{j=0}^h \sum_{i=0}^j \rho_g^i e_{t+j-i}^g \right].$$

We can compute the values by plugging in the parameter values. Figure 1 shows that plot.

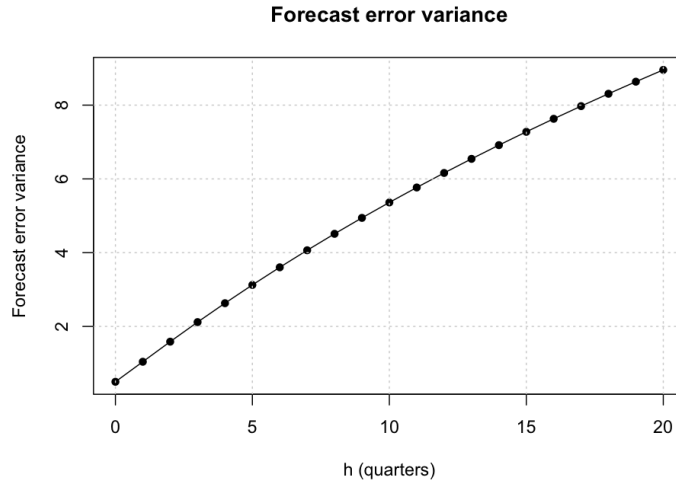


Figure 1: Forecast Error Variance

c. The decomposition is shown in Figure 2.

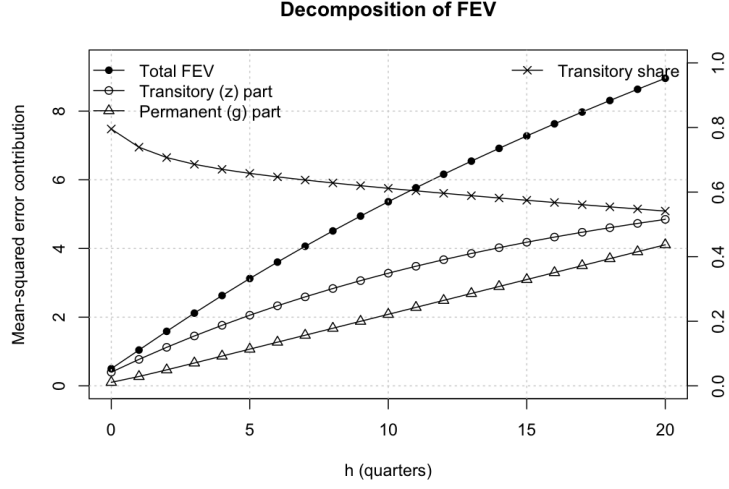


Figure 2: Decomposition of Forecast Error Variance

The effect of transitory components decays with time. The permanent components have a more persistent effect. From the expressions of FEV^z and FEV^g , mathematically, we can observe that each term of e^z is summed once, while each term of e^g is summed multiple times. This makes the effect from permanent components accumulate.

Question 2

a. Rewriting the $ARMA(1, 1)$ process, we get

$$(1 - 0.8L)y_t = (1 + 0.4L)e_t.$$

Suppose that $y_t = \psi(L)e_t$. Then we have

$$(1 - 0.8L)(\psi_0 + \psi_1L + \cdots) = 1 + 0.4L.$$

Equating coefficients, we get the impulse responses:

$$\psi_j = \begin{cases} 1, & j = 0 \\ 1.2 \times 0.8^{j-1}, & j \geq 1 \end{cases}$$

Plotting this function, we get Figure 3. Codes in “hw2-q2.R”.

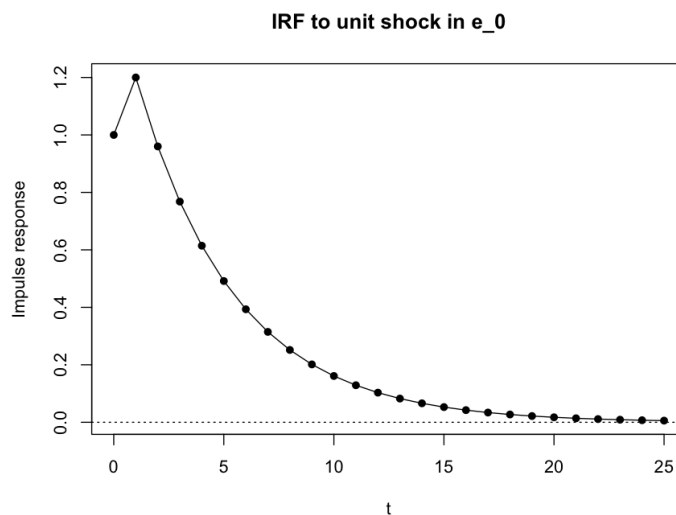


Figure 3: IRF to unit shock in e_0

b. We can write y_t as a $MA(\infty)$ process:

$$y_t = \sum_{j=0}^{\infty} \psi_j e_{t-j}.$$

It immediately follows that $\mathbb{E}[y_t] = 0$.

The variance is

$$\text{Var}[y_t] = \text{Var}[e_t] \left(\sum_{j=0}^{\infty} \psi_j^2 \right) = 3 \left(1 + \frac{1.2^2}{1 - 0.64} \right) = 15.$$

The autocovariances are

$$\gamma(1) = \text{Var}[e_t] \left(\sum_{j=0}^{\infty} \psi_j \psi_{j+1} \right) = 3 \left(1.2 + \frac{1.2^2 \times 0.8}{1 - 0.64} \right) = 13.2.$$

$$\gamma(2) = \text{Var}[e_t] \left(\sum_{j=0}^{\infty} \psi_j \psi_{j+2} \right) = 3 \left(0.96 + \frac{1.2^2 \times 0.8^2}{1 - 0.64} \right) = 10.56.$$

c. Since $\text{Var}[e_t] = 3$, we have the ACGF $G : \mathbb{C} \rightarrow \mathbb{C}$ defined as

$$\begin{aligned} G(z) &= 3 \left(1 + \sum_{j=1}^{\infty} 1.2 \times 0.8^{j-1} z^j \right) \left(1 + \sum_{j=1}^{\infty} 1.2 \times 0.8^{j-1} z^{-j} \right) \\ &= 3 \times \left[\left(1 + 1.44 \times \sum_{j=1}^{\infty} 0.8^{2(j-1)} \right) + \left(1.2 + 1.44 \times \sum_{j=1}^{\infty} 0.8^{2j-1} \right) z \right. \\ &\quad \left. + \left(0.96 + 1.44 \times \sum_{j=1}^{\infty} 0.8^{2j} \right) z^2 + \dots \right]. \end{aligned}$$

By the ACGF,

$$\gamma(0) = 15, \gamma(1) = 13.2, \gamma(2) = 10.56.$$

This verifies the results in part (b).

- d. I used $e \sim \mathcal{N}(0, 3)$ to generate the innovations. The figure 4 is the simulated path of y . Codes in “hw2_q2.R”.

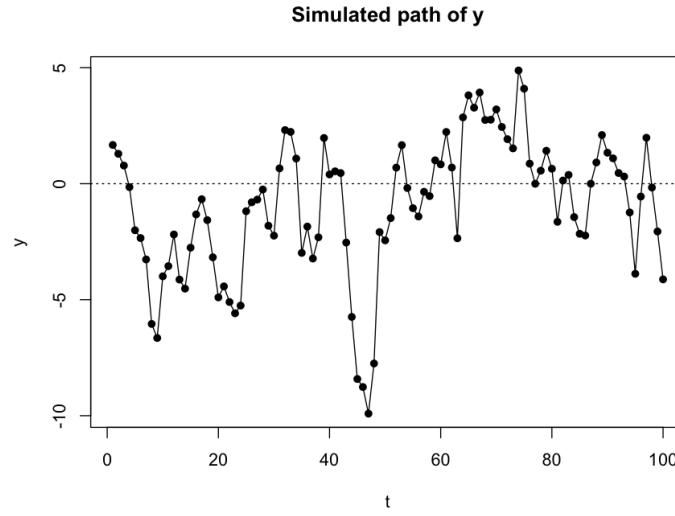


Figure 4: Simulated path of y

The reason for burning is to gain stationarity in the long run.

- e. i. I tried $M = 3$ in the algorithm. The following tables are the reports:

	Estimate	Std. Error	t value	$\Pr(> t)$
(Intercept)	-0.0154	0.0093	-1.65	0.1024
y_lag	0.6891	0.0035	196.14	0.0000
e_lag	0.3882	0.0063	61.57	0.0000

Table 1: Hannan–Rissanen Estimates

- ii. The MoM estimates are $\hat{\alpha} = 0.6785974$ and $\hat{\theta} = 0.4163043$.

f. Table 2 summarizes the simulation results.

method	parameter	true	mean_hat	bias	mse	sd
HR	alpha	0.80	0.76	-0.04	0.01	0.08
HR	theta	0.40	0.40	-0.00	0.01	0.11
MoM	alpha	0.80	0.78	-0.02	0.01	0.08
MoM	theta	0.40	0.44	0.04	0.04	0.19

Table 2: Parameter estimates statistics.

Distributions are in Figure 5.

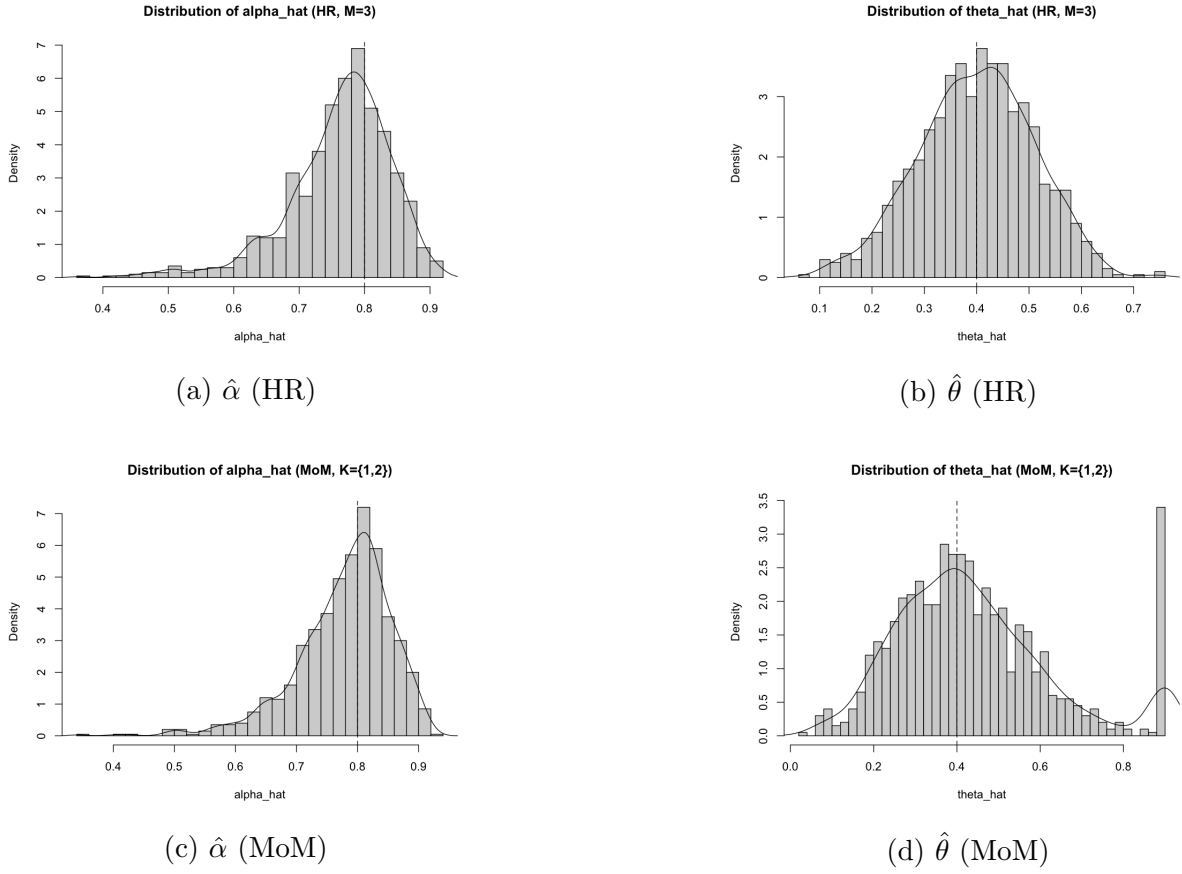


Figure 5: Distributions of parameter estimates. Dashed vertical line indicates the true parameter value.

g. First, $T = 50$.

Table 3 summarizes the simulation results.

method	parameter	true	mean_hat	bias	mse	sd
HR	alpha	0.80	0.71	-0.09	0.02	0.13
HR	theta	0.40	0.41	0.01	0.02	0.15
MoM	alpha	0.80	0.75	-0.05	0.02	0.12
MoM	theta	0.40	0.46	0.06	0.06	0.24

Table 3: Parameter estimates statistics.

Distributions are in Figure 6.

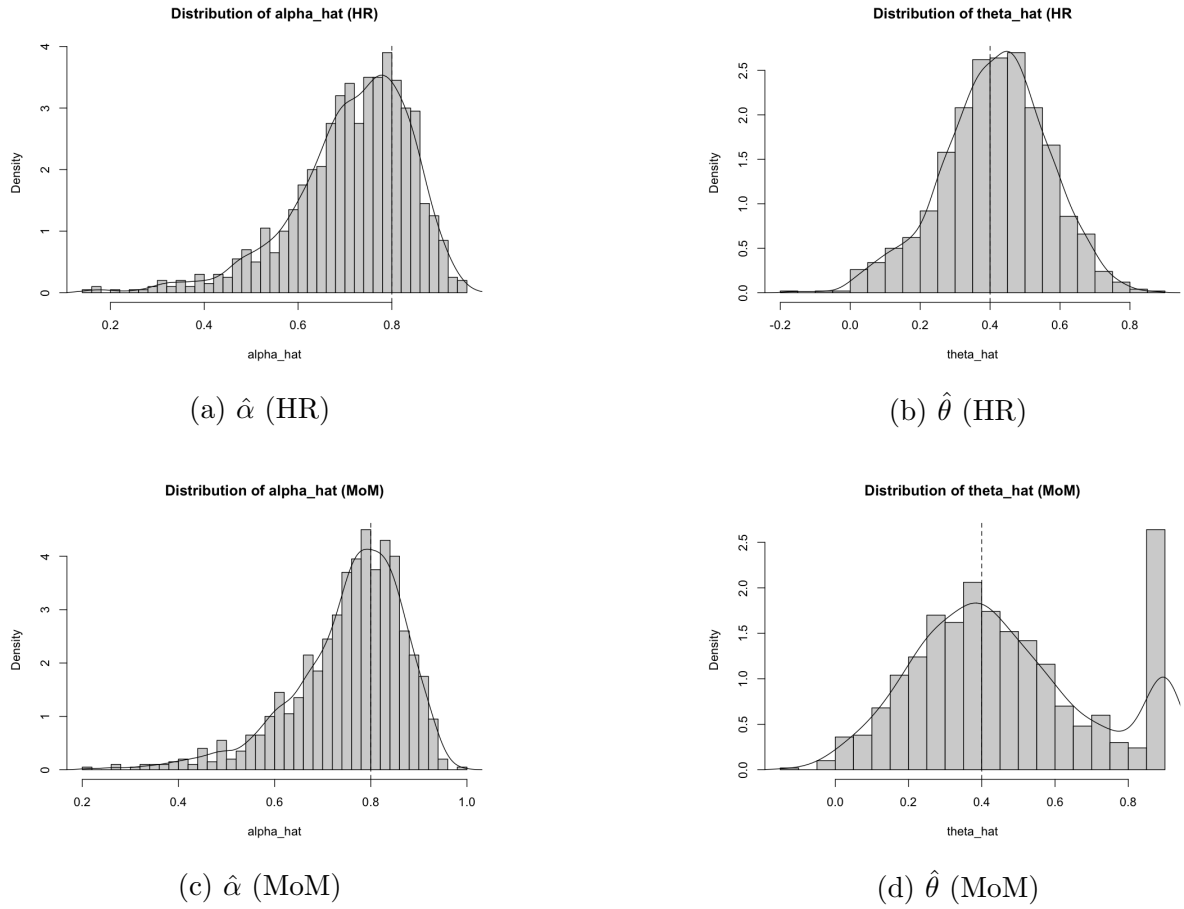


Figure 6: Distributions of parameter estimates. Dashed vertical line indicates the true parameter value.

Then $T = 500$.

Table 4 summarizes the simulation results.

method	parameter	true	mean_hat	bias	mse	sd
HR	alpha	0.80	0.79	-0.01	0.00	0.03
HR	theta	0.40	0.40	-0.00	0.00	0.05
MoM	alpha	0.80	0.79	-0.01	0.00	0.03
MoM	theta	0.40	0.41	0.01	0.00	0.07

Table 4: Parameter estimates statistics.

Distributions are in Figure 7.

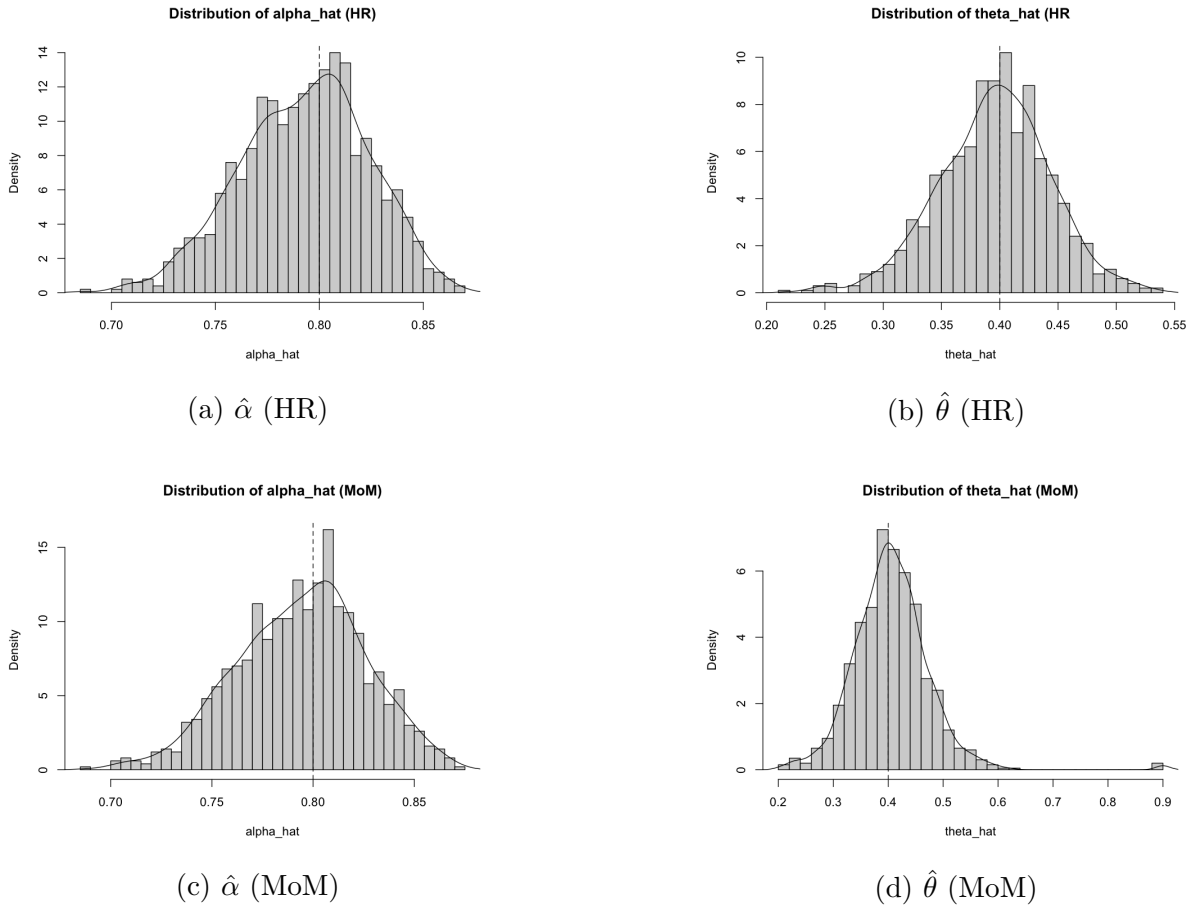


Figure 7: Distributions of parameter estimates. Dashed vertical line indicates the true parameter value.

- h. As T increases, the mean value of the estimates converges to the true parameter value, the MSE vanishes, and the standard deviation decreases. The distribution are more concentrated around the true parameter value.

I also observe a spike around 1 in the distribution of $\hat{\theta}$ using MoM when T is small. The density at (around) 1 shrinks as T increases. This is because the sample is noisy when the size is small so that the MSE function is flatter and the optimizer will fall around the boundary. The identification of θ is strengthened when the sample size becomes larger.

Question 3

a. Figure 8 shows the results.

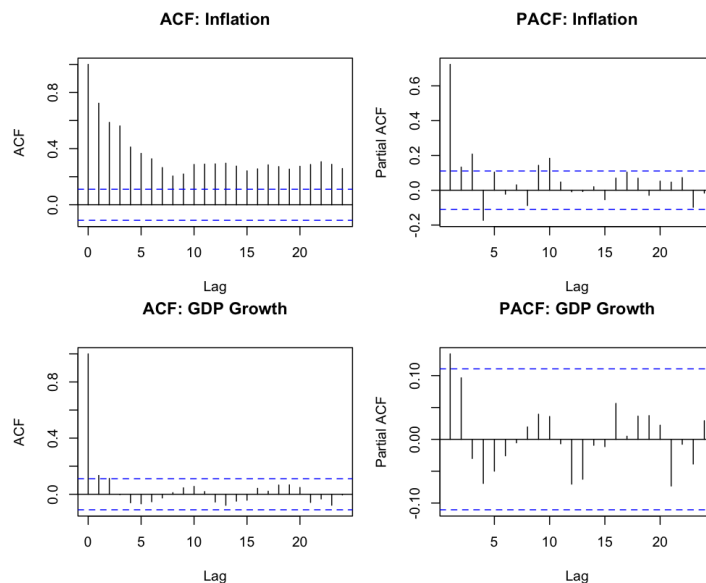


Figure 8: ACF and PACF of Inflation and GDP Growth

Inflation shows strong positive autocorrelation while GDP growth does not. Both series have PACF dying out quickly, while the partial serial correlations of the two time series at similar lags are of the same sign, in general.

b. The Box-Jenkins principle says that we should use the smallest number of parameters for adequate representation.

For inflation, the serial correlation is quite persistent. The PACF dies out after lag 2. I would pick $ARMA(3, 2)$ to account for the persistence.

For GDP growth, both ACF and PACF dies out at around lag 1. So I would pick $ARMA(1, 1)$.

- c. Table 5 shows the BIC values for AR(p) models, $p=0, \dots, 10$. For inflation, $p = 4$ gives us the lowest BIC and for GDP growth $p = 0$ gives us the lowest BIC. So we choose $AR(4)$ for inflation and no autocorrelation for GDP growth.

Table 5: BIC values for AR(p) models, $p=0, \dots, 10$

p	Inflation	GDPGrowth
0	761.06	968.23
1	534.32	968.29
2	534.05	971.08
3	524.68	976.53
4	521.16	980.74
5	523.72	985.69
6	529.32	991.24
7	535.03	996.98
8	537.59	1002.58
9	536.92	1007.80
10	530.51	1013.15

d. Test statistics are in Table 6.

Table 6: Diagnostic Tests: Ljung–Box and Jarque–Bera

Series	LB	p -value	JB	p -value
Inflation (AR(4))	24.27	0.00	1176.27	0.00
GDP Growth (AR(0))	16.38	0.17	3342.39	0.00

The $AR(4)$ model for inflation is not enough to capture serial correlation because the p -value of Ljung-Box still shows that there exists autocorrelation in the residuals. The Jarque-Bera test shows that the residuals are not normally distributed.

The $AR(0)$ model for GDP growth seems appropriate in the sense that the p -value of the Ljung-Box test is large. However, the p -value for the Jarque-Bera test is still small, indicating that the residuals are not normally distributed.

The ACF and PACF of the residuals are in Figure 9.

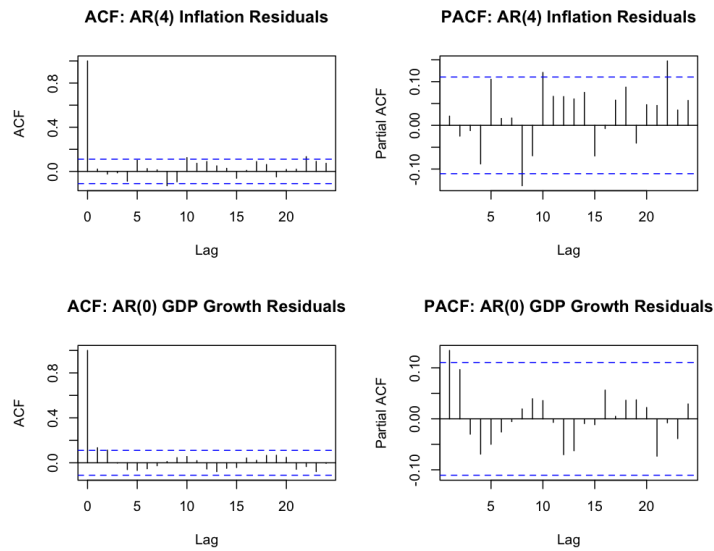


Figure 9: ACF and PACF of Residuals

If the true DGP is truly $AR(p)$, then the ACF and the PACF of the residuals should be both (approximately) zero at all lags. However, the $AR(4)$ for inflation fails this test while $AR(0)$ for GDP growth (almost) passes this test.

The above diagnostic shows that the true DGP may not be a simple $AR(p)$ process for inflation, while $AR(0)$ may be a correct one for GDP growth rate.

e. Fan charts are in Figure 10.

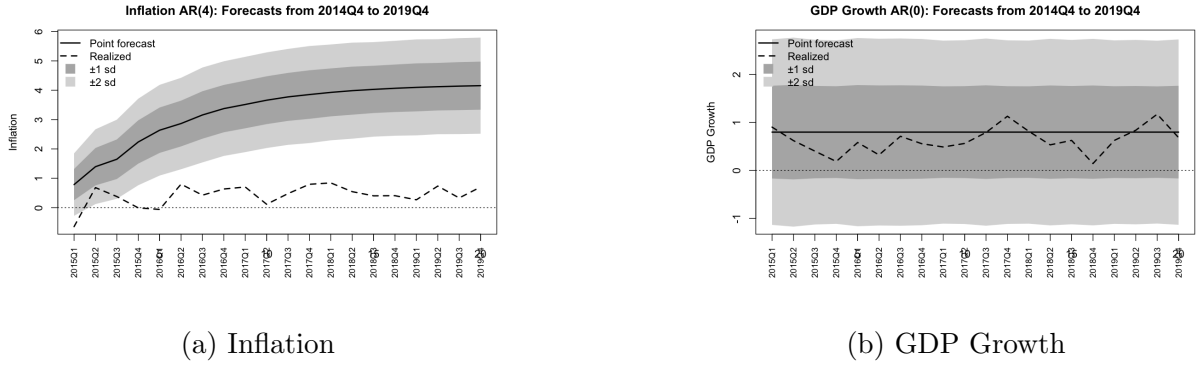


Figure 10: Fan Charts.

The forecast errors are in Figure 11.

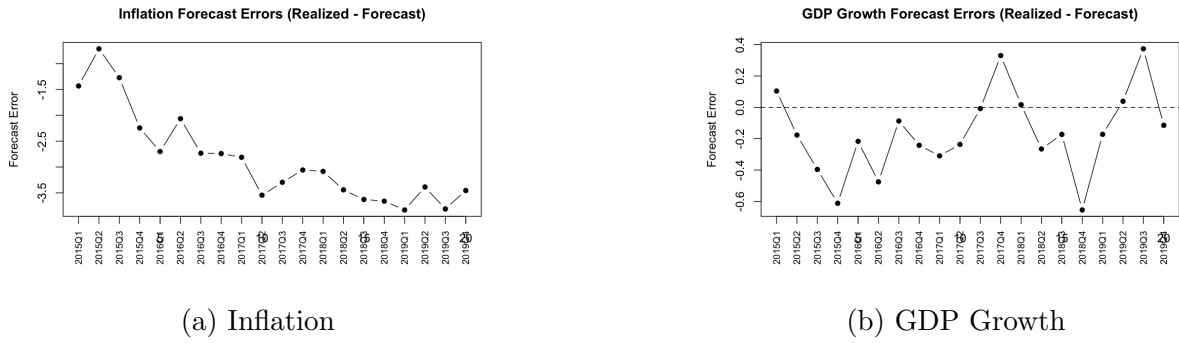


Figure 11: Forecast Errors.

It is clear that the $AR(4)$ model for inflation is not able to well predict the inflation rate since the difference is quite large in magnitude. The prediction of GDP growth rate using mean ($AR(0)$) is more reasonable. The difference is much smaller in magnitude. The comparison echoes with the diagnostics in part (d).

Question 4

a. The expectation is

$$\begin{aligned}\mathbb{E}[x_t] &= \mathbb{E}[\alpha] \cos t + \mathbb{E}[\beta] \sin t \\ &= 0.\end{aligned}$$

The variance is

$$\begin{aligned}\text{Var}[x_t] &= \text{Var}[\alpha] \cos^2 t + \text{Var}[\beta] \sin^2 t + 2 \text{Cov}(\alpha, \beta) \cos t \sin t \\ &= \cos^2 t + \sin^2 t \\ &= 1.\end{aligned}$$

The second line is by the independence of α and β .

b. No. Note that

$$\begin{aligned}\mathbb{E}[\alpha^2] &= \text{Var}[\alpha] + (\mathbb{E}[\alpha])^2 \\ &= 1 \\ \mathbb{E}[\beta^2] &= \text{Var}[\beta] + (\mathbb{E}[\beta])^2 \\ &= 1 \\ \mathbb{E}[\alpha\beta] &= \text{Cov}(\alpha, \beta) + \mathbb{E}[\alpha]\mathbb{E}[\beta] \\ &= 0.\end{aligned}$$

As a result, we have

$$\begin{aligned}\mathbb{E}[x_t x_{t-k}] &= \mathbb{E}[\alpha^2] \cos t \cos(t-k) + \mathbb{E}[\beta^2] \sin t \sin(t-k) \\ &\quad + \mathbb{E}[\alpha\beta][\cos t \sin(t-k) + \sin t \cos(t-k)] \\ &= \cos t \cos(t-k) + \sin t \sin(t-k) \\ &= \cos[t - (t-k)] \\ &= \cos k.\end{aligned}$$